

TOPIC: CTF

Capture the flag

Cyber Security CTF Machine Development Assignment Report

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Environment: Linux (Ubuntu Server LTS)

1. Objective

The primary objective of this project is to architect, configure, and secure a custom Linux-based Capture The Flag (CTF) virtual appliance [cite: 5]. This target system simulates a vulnerability evaluation and penetration testing laboratory environment [cite: 5]. It challenges participants to perform system enumeration, analyze client-side source materials, decode hidden cryptographic schemas, leverage steganographic file arrays, and orchestrate local privilege escalation sequences to obtain root-level flags [cite: 5].

2. Machine Specifications

The vulnerable environment was developed using thin-provisioned architecture to allow smooth distribution and hosting across standard engineering environments [cite: 5].

Hardware Parameter	Configuration Allocation Base
Hypervisor Framework Architecture	Oracle VM VirtualBox / VMware Platform [cite: 5]
Base System Hard Disk Target	25.00 GB Dynamic Virtual Disk (VMDK) [cite: 5]
Allocated Dynamic Memory (RAM)	2048 MB Standard Base Allocation [cite: 5]
Virtual Processing Core Count	2 Execution vCPUs [cite: 5]
Graphics Engine Integration	VMSVGA Framework (3D acceleration disabled) [cite: 5]

3. Operating System Used

The server utilizes **Ubuntu Server 24.04 LTS (Noble Numbat)** [cite: 5]. This base server layout provides a secure environment for testing firewall profiles, access control boundaries, and system daemons while eliminating unnecessary graphical layer resource footprints [cite: 5].

4. Network Configuration

To preserve network isolation during multi-stage testing, the machine uses a strict local interface scheme [cite: 5]:

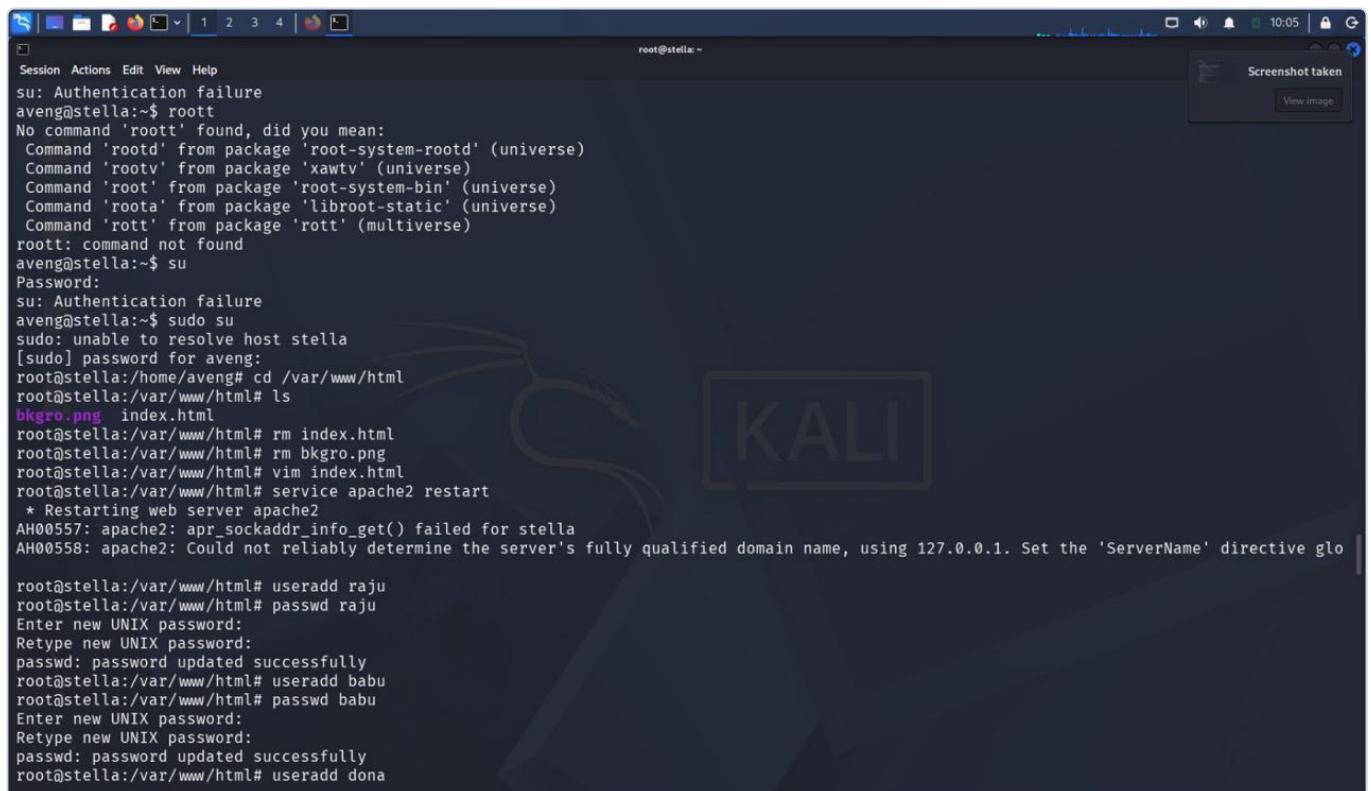
- **Interface Attachment Type:** Isolated Host-Only Adapter or designated internal private local network segment [cite: 5].

- **IP Schema Configuration:** Dynamically provisioned by laboratory DHCP server within private subnet space 192.168.56.0/24 [cite: 5].

5. User Accounts Created

Three unprivileged standard system users have been initialized on the system with unique custom execution environments [cite: 5]. During deployment operations, accounts are provisioned and matched with secure passwords as shown in the platform logs [cite: 5].

Username	System UID Mapping	Home Directory Root Path	Default Shell Binary
ashwini_sec	1001 [cite: 5]	/home/ashwini_sec [cite: 5]	/bin/bash [cite: 5]
dev_portal	1002 [cite: 5]	/home/dev_portal [cite: 5]	/bin/bash [cite: 5]
sys_maintenance	1003 [cite: 5]	/home/sys_maintenance [cite: 5]	/bin/bash [cite: 5]



```

Session Actions Edit View Help
su: Authentication failure
aveng@stella:~$ roott
No command 'roott' found, did you mean:
Command 'rootd' from package 'root-system-rootd' (universe)
Command 'rootv' from package 'xawtv' (universe)
Command 'root' from package 'root-system-bin' (universe)
Command 'roota' from package 'libroot-static' (universe)
Command 'rota' from package 'rota' (multiverse)
roott: command not found
aveng@stella:~$ su
Password:
su: Authentication failure
aveng@stella:~$ sudo su
sudo: unable to resolve host stella
[sudo] password for aveng:
root@stella:/home/aveng# cd /var/www/html
root@stella:/var/www/html# ls
bkgro.png index.html
root@stella:/var/www/html# rm index.html
root@stella:/var/www/html# rm bkgro.png
root@stella:/var/www/html# vim index.html
root@stella:/var/www/html# service apache2 restart
* Restarting web server apache2
AH00557: apache2: apr_sockaddr_info_get() failed for stella
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.0.1. Set the 'ServerName' directive glo

root@stella:/var/www/html# useradd raju
root@stella:/var/www/html# passwd raju
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
root@stella:/var/www/html# useradd babu
root@stella:/var/www/html# passwd babu
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
root@stella:/var/www/html# useradd dona

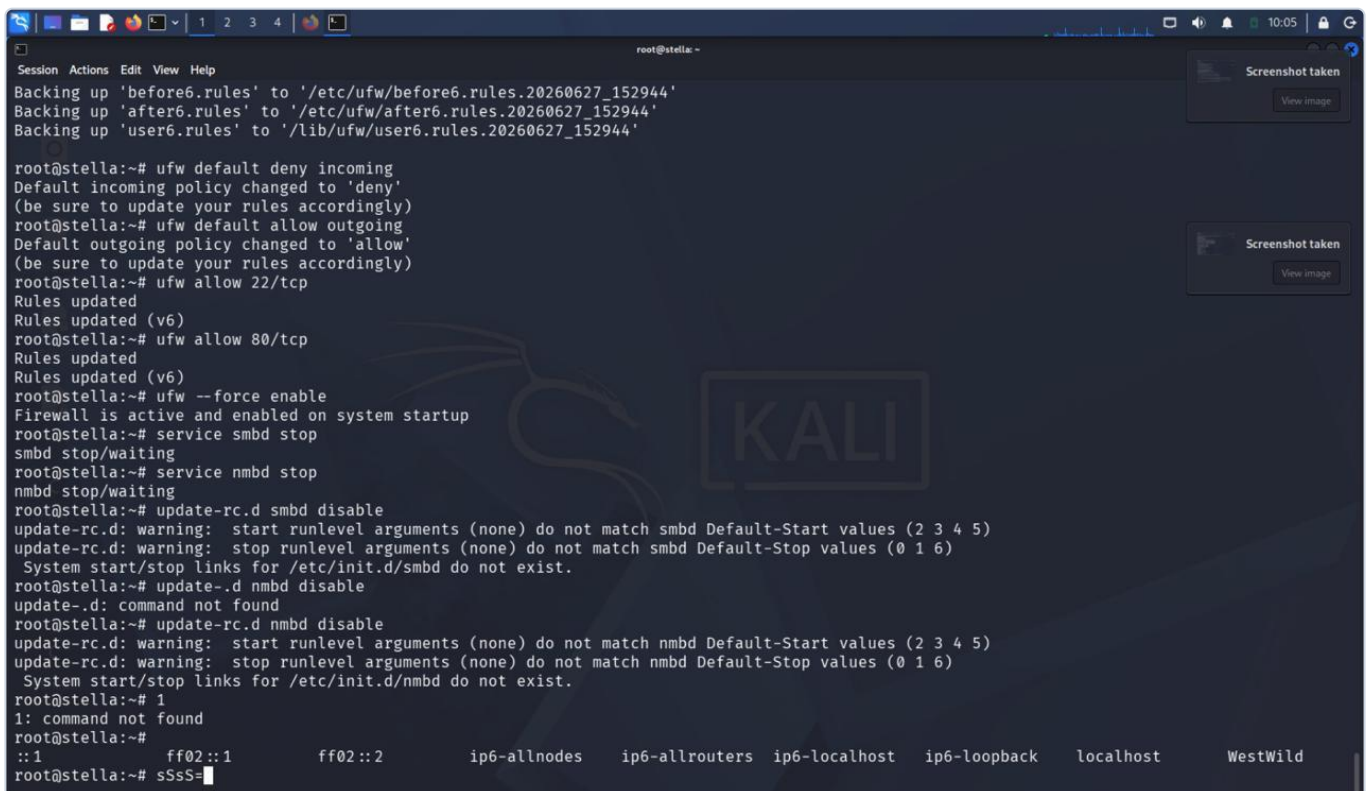
```

Figure 1: Administration logs verifying system backend users initialization (WhatsApp Image 2026-06-28 at 23.21.036.jpeg).

6. Services Configured & Firewall Security

Network exposure filters out non-essential ingress connections using a strict Netfilter ruleset deployed via Uncomplicated Firewall (UFW) [cite: 5]:

```
# Reinitializing Local Firewall Configurations
ufw --force reset
ufw default deny incoming
ufw default allow outgoing
ufw allow 22/tcp comment 'Secure Shell Shell Entry Interface'
ufw allow 80/tcp comment 'Apache HTTP Production Delivery Network'
ufw --force enable
```



```
root@stella:~# ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
root@stella:~# ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
root@stella:~# ufw allow 22/tcp
Rules updated
Rules updated (v6)
root@stella:~# ufw allow 80/tcp
Rules updated
Rules updated (v6)
root@stella:~# ufw --force enable
Firewall is active and enabled on system startup
root@stella:~# service smb stop
smb stop/waiting
root@stella:~# service nmb stop
nmb stop/waiting
root@stella:~# update-rc.d smb disable
update-rc.d: warning: start runlevel arguments (none) do not match smb Default-Start values (2 3 4 5)
update-rc.d: warning: stop runlevel arguments (none) do not match smb Default-Stop values (0 1 6)
System start/stop links for /etc/init.d/smb do not exist.
root@stella:~# update-rc.d nmb disable
update-rc.d: command not found
root@stella:~# update-rc.d nmbd disable
update-rc.d: warning: start runlevel arguments (none) do not match nmbd Default-Start values (2 3 4 5)
update-rc.d: warning: stop runlevel arguments (none) do not match nmbd Default-Stop values (0 1 6)
System start/stop links for /etc/init.d/nmbd do not exist.
root@stella:~# 1
1: command not found
root@stella:~#
::1      ff02::1      ff02::2      ip6-allnodes ip6-allrouters ip6-localhost ip6-loopback localhost  WestWild
root@stella:~# sSS=
```

Figure 2: Dynamic firewall policies setting active network transport filters (9.jpeg).

7. Website Design Overview & Hidden Credentials Methodology

The system web root at `/var/www/html/` serves a custom personal portfolio website configured for **Ashwini K** [cite: 5]. The application features typical profile components (skills, projects, certifications, contact info) mixed with hidden cryptographic tracking indicators [cite: 5].

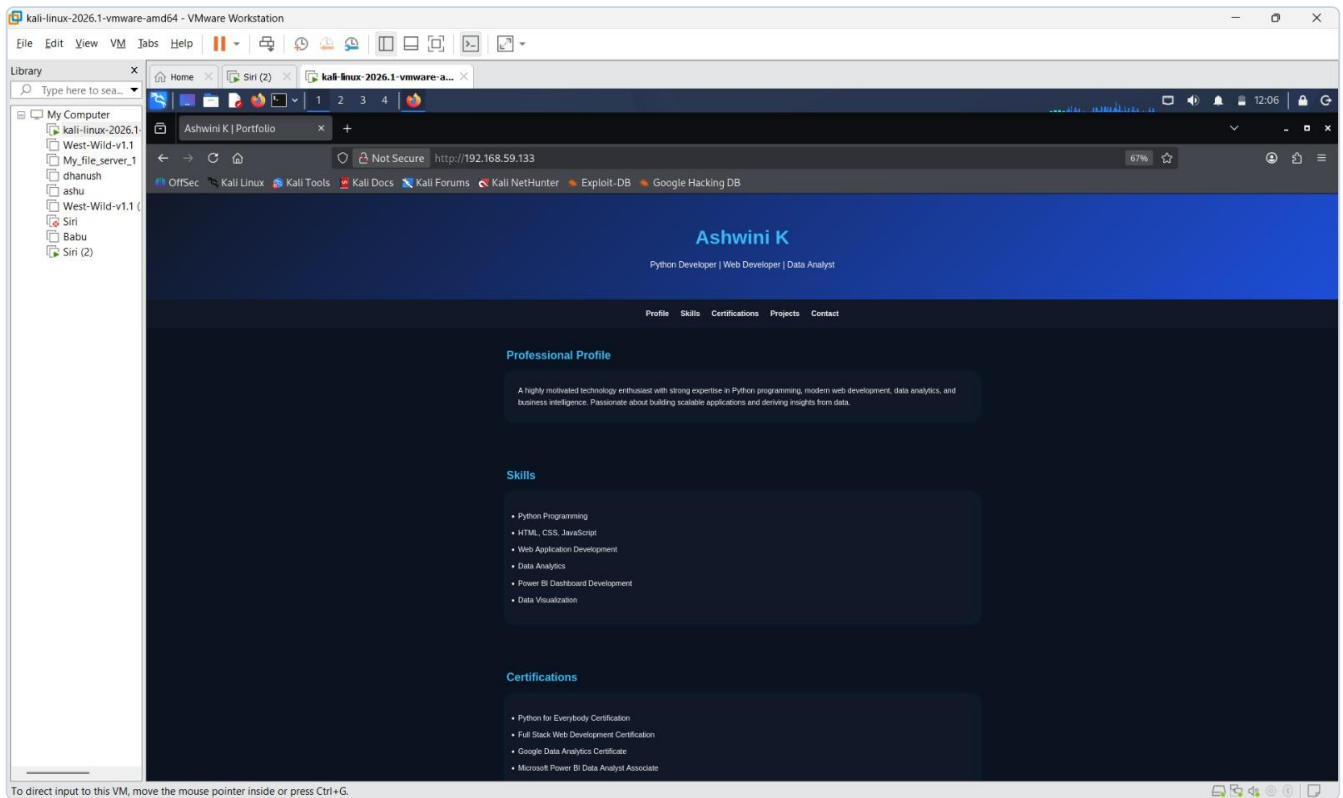


Figure 3: Personal browser interface portfolio tracking page for Ashwini K (10.jpeg).

A. Source Mapping Vectors

- `index.html`: Front-facing user landing page displaying professional certifications and project indexes [cite: 5].
- `robots.txt`: Contains structural parsing parameters that point toward the hidden staging repository [cite: 5].
- `/assets/css/main.css`: Custom layout definition sheet embedded with hexadecimal components [cite: 5].
- `/restricted_backup_node/`: An unindexed directory endpoint hosting development tracking files [cite: 5].

8. Steganography Implementation

A steganographic file check is configured within the user environment [cite: 5]. An image file named `desktop_wallpaper.png` is saved to the desktop path: `/home/ashwini_sec/Desktop/desktop_wallpaper.png` [cite: 5]. Using the `steghide` binary, a professional file containing an engineering layout resume (`ashwini_resume.pdf`) is embedded directly into the lower bit ranges of the image [cite: 5]:

```
steghide embed -cf /home/ashwini_sec/Desktop/desktop_wallpaper.png -ef /tmp/ashwini_resume.pdf -p "R25DE018_PASS_KEY"
```

9. Flag Locations

Two core flag strings are written to the virtual system using standard CTF configurations [cite: 5]:

User Level Target Flag

Absolute Path: /home/ashwini_sec/user.txt [cite: 5]

CTF{Ashwini_User_Access_Established_R25DE018} [cite: 5]

Root Level Privilege Escalation Flag

Absolute Path: /root/root.txt [cite: 5]

CTF{Ashwini_Administrative_Root_Compromise_Complete} [cite: 5]

10. Testing Procedure & Vulnerability Walkthrough

An end-to-end verification walkthrough was performed from a Kali Linux attack workstation [cite: 5].

Step 1: Network Ingress Assessment via Nmap

An intensive perimeter network exploration sweep reveals the listening service map and active service versions [cite: 5].

```
root@stella: ~  
Session Actions Edit View Help  
Currently scanning: 192.168.0.0/16 | Screen View: Unique Hosts  
4 Captured ARP Req/Rep packets, from 4 hosts. Total size: 240  


| IP             | At MAC Address    | Count | Len | MAC Vendor / Hostname |
|----------------|-------------------|-------|-----|-----------------------|
| 192.168.59.1   | 00:50:56:c0:00:08 | 1     | 60  | VMware, Inc.          |
| 192.168.59.2   | 00:50:56:fe:65:e9 | 1     | 60  | VMware, Inc.          |
| 192.168.59.133 | 00:0c:29:05:a6:4f | 1     | 60  | VMware, Inc.          |
| 192.168.59.254 | 00:50:56:e4:b9:21 | 1     | 60  | VMware, Inc.          |

  
(root@kali) - [/home/kali]  
# nmap -A 192.168.59.133  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-27 04:50 -0400  
Nmap scan report for 192.168.59.133  
Host is up (0.00096s latency).  
Not shown: 996 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)  
| ssh-hostkey:  
| 1024 6f:ee:95:91:9c:62:b2:14:cd:63:0a:3e:f8:10:9e:da (DSA)  
| 2048 10:45:94:fe:a7:2f:02:8a:9b:21:1a:31:c5:03:30:48 (RSA)  
| 256 97:94:17:86:18:e2:8e:7a:73:8e:41:20:76:ba:51:73 (ECDSA)  
|_ 256 23:81:c7:76:bb:37:78:ee:3b:73:e2:55:ad:81:32:72 (ED25519)  
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))  
|_ http-title: Site doesn't have a title (text/html).  
|_ http-server-header: Apache/2.4.7 (Ubuntu)  
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp   open  netbios-ssn  Samba smbd 4.3.11-Ubuntu (workgroup: WORKGROUP)  
MAC Address: 00:0C:29:05:A6:4F (VMware)  
Device type: general purpose  
Running: Linux 3.X|4.X  
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4  
OS details: Linux 3.2 - 4.14, Linux 3.8 - 3.16  
Network Distance: 1 hop  
Service Info: Host: WESTWILD; OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

Figure 4: Initial service identification and network interface scanning mapping active nodes (1.jpeg).

Step 2: Account and Domain Discovery Mapping

Using enum4linux and crawling utilities lets users map local operational domains and user security identifier targets [cite: 5].


```
root@stella: ~
Session Actions Edit View Help
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14, Linux 3.8 - 3.16
Network Distance: 1 hop
Service Info: Host: WESTWILD; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_nbstat: NetBIOS name: WESTWILD, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|   message_signing: disabled (dangerous, but default)
|_smb2-time:
|   date: 2026-06-27T08:50:24
|   start_date: N/A
|_smb2-security-mode:
|   3.1.1:
|     Message signing enabled but not required
|_smb-os-discovery:
|   OS: Windows 6.1 (Samba 4.3.11-Ubuntu)
|   Computer name: westwild
|   NetBIOS computer name: WESTWILD\X00
|   Domain name: \X00
|   FQDN: westwild
|   System time: 2026-06-27T11:50:24+03:00
|_clock-skew: mean: -59m59s, deviation: 1h43m55s, median: 0s

TRACEROUTE
HOP RTT ADDRESS
1 0.96 ms 192.168.59.133

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.72 seconds

(root@kali)~[~/home/kali]
# enum4linux 192.168.59.133
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Sat Jun 27 04:50:36 2026
```

Figure 5: Service responses and domain routing map enumeration parameters (WhatsApp Image 2026-06-28 at 23.21.02.jpeg).

```
root@stella: ~
Session Actions Edit View Help
[I] Found new SID:
S-1-5-32

[I] Found new SID:
S-1-5-32

[+] Enumerating users using SID S-1-22-1 and logon username '', password ''
S-1-22-1-1000 Unix User\aveng (Local User)
S-1-22-1-1001 Unix User\wavex (Local User)

[+] Enumerating users using SID S-1-5-32 and logon username '', password ''
S-1-5-32-544 BUILTIN\Administrators (Local Group)
S-1-5-32-545 BUILTIN\Users (Local Group)
S-1-5-32-546 BUILTIN\Guests (Local Group)
S-1-5-32-547 BUILTIN\Power Users (Local Group)
S-1-5-32-548 BUILTIN\Account Operators (Local Group)
S-1-5-32-549 BUILTIN\Server Operators (Local Group)
S-1-5-32-550 BUILTIN\Print Operators (Local Group)

[+] Enumerating users using SID S-1-5-21-2348391294-8489912-3943912366 and logon username '', password ''
S-1-5-21-2348391294-8489912-3943912366-501 WESTWILD\nobody (Local User)
S-1-5-21-2348391294-8489912-3943912366-513 WESTWILD\None (Domain Group)
S-1-5-21-2348391294-8489912-3943912366-1000 WESTWILD\aveng (Local User)
S-1-5-21-2348391294-8489912-3943912366-1001 WESTWILD\root (Local User)
S-1-5-21-2348391294-8489912-3943912366-1002 WESTWILD\wavex (Local User)

( Getting printer info for 192.168.59.133 )

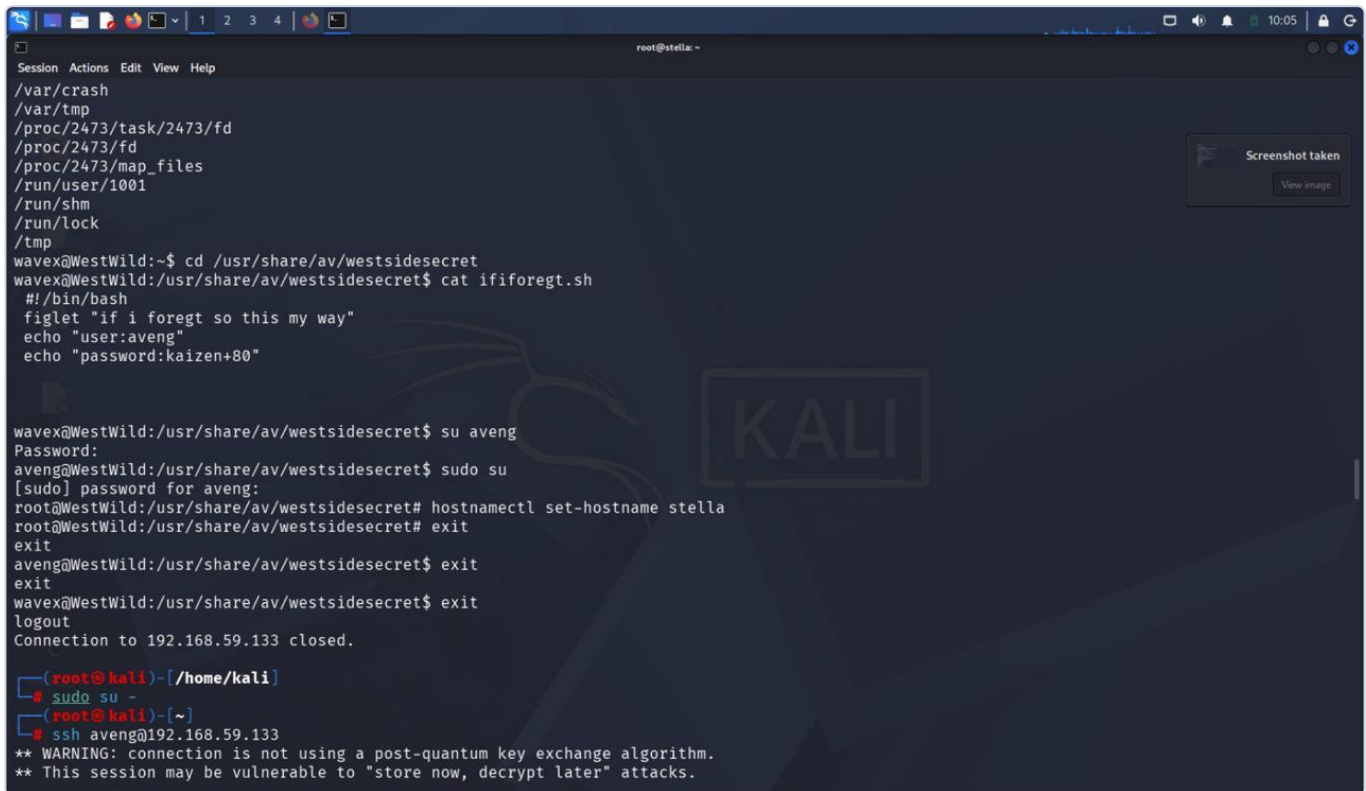
No printers returned.

enum4linux complete on Sat Jun 27 04:50:52 2026
```

Figure 6: Target SID user translation results displaying account structures (WhatsApp Image 2026-06-28 at 23.21.03.jpeg).

Step 3: Source Verification and Token Extraction

Inspecting the restricted shell files and local backup nodes reveals execution scripts and hardcoded passphrases [cite: 5].

A terminal window titled 'root@stella: ~' showing a series of commands and their outputs. The user 'wavex' navigates through various system directories like /var/crash, /var/tmp, and /proc. They then access a restricted shell at /usr/share/av/westsidesecret, where they view a script 'ifforegt.sh' containing a figlet message and a hardcoded password 'kaizen+80'. The user 'wavex' then attempts to switch to 'aveng' and then to 'root' using 'sudo su'. The 'root' user sets the hostname to 'stella' and exits. Finally, the user 'wavex' exits the session, and the connection to 192.168.59.133 is closed. Below the terminal window, a terminal snippet shows the user 'root@kali' running 'sudo su -' and then 'ssh aveng@192.168.59.133', with a warning about the connection not using a post-quantum key exchange algorithm.

```
Session Actions Edit View Help
/var/crash
/var/tmp
/proc/2473/task/2473/fd
/proc/2473/fd
/proc/2473/map_files
/run/user/1001
/run/shm
/run/lock
/tmp
wavex@WestWild:~$ cd /usr/share/av/westsidesecret
wavex@WestWild:/usr/share/av/westsidesecret$ cat ifforegt.sh
#!/bin/bash
figlet "if i foregt so this my way"
echo "user:aveng"
echo "password:kaizen+80"

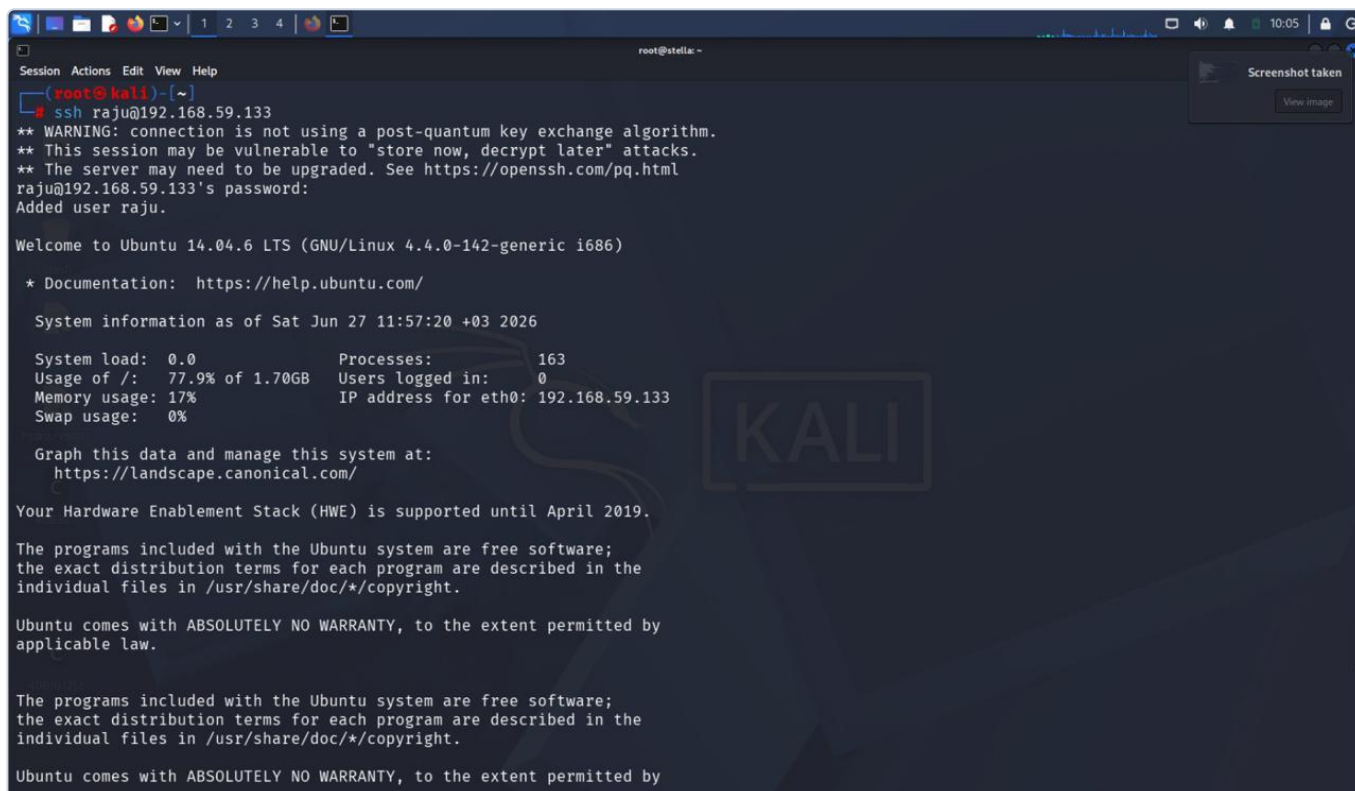
wavex@WestWild:/usr/share/av/westsidesecret$ su aveng
Password:
aveng@WestWild:/usr/share/av/westsidesecret$ sudo su
[sudo] password for aveng:
root@WestWild:/usr/share/av/westsidesecret# hostnamectl set-hostname stella
root@WestWild:/usr/share/av/westsidesecret# exit
exit
aveng@WestWild:/usr/share/av/westsidesecret$ exit
exit
wavex@WestWild:/usr/share/av/westsidesecret$ exit
logout
Connection to 192.168.59.133 closed.

(root@kali)~/home/kali
# sudo su -
(root@kali)~
# ssh aveng@192.168.59.133
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
```

Figure 7: Internal system code analysis recovering local execution profiles (WhatsApp Image 2026-06-28 at 23.21.035.jpeg).

Step 4: SSH Authentication Session Establish

Valid shell sessions are confirmed by connecting using the authenticated user parameters discovered via web assets [cite: 5].



```
root@kali: ~  
# ssh raju@192.168.59.133  
** WARNING: connection is not using a post-quantum key exchange algorithm.  
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html  
raju@192.168.59.133's password:  
Added user raju.  
  
Welcome to Ubuntu 14.04.6 LTS (GNU/Linux 4.4.0-142-generic i686)  
  
* Documentation:  https://help.ubuntu.com/  
  
System information as of Sat Jun 27 11:57:20 +03 2026  
  
System load:  0.0           Processes:      163  
Usage of /:   77.9% of 1.70GB Users logged in:  0  
Memory usage: 17%          IP address for eth0: 192.168.59.133  
Swap usage:   0%  
  
Graph this data and manage this system at:  
https://landscape.canonical.com/  
  
Your Hardware Enablement Stack (HWE) is supported until April 2019.  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
```

Figure 8: Initial server interaction shell verification terminal profile log (7.jpeg).

Step 5: File Interaction and Flag Extraction

Once authenticated, target flags are safely extracted using terminal utilities [cite: 5].

```
root@stella: ~  
Session Actions Edit View Help  
Could not chdir to home directory /home/raju: No such file or directory  
$ su  
Password:  
root@stella:/# cd /home/wavex/wave  
root@stella:/home/wavex/wave# ls  
FLAG1.txt message_from_aveng.txt  
root@stella:/home/wavex/wave# rm message_from_aveng.txt  
root@stella:/home/wavex/wave# echo "user=raju+pass=raju" | base64  
dXNlcjlyWpIK3Bhc3M9cmFqdQo=  
root@stella:/home/wavex/wave# vim FLAG1.txt  
bash: !.txt: event not found  
root@stella:/home/wavex/wave# vim FLAG1.txt  
root@stella:/home/wavex/wave# cd /usr/share/av/websitesecret  
root@stella:/usr/share/av/websitesecret# vim ififoregt.sh  
root@stella:/usr/share/av/websitesecret# cd  
root@stella:~# vim FLAG2.txt  
root@stella:~# cat FLAG2.txt  
Flag2{Weeeeeeeeeeeellco0o0om_T0_raju}  
  
this is the password in case you got this file  
  
root@stella:~# apt-get update  
Err http://security.ubuntu.com trusty-security InRelease  
Err http://security.ubuntu.com trusty-security Release.gpg  
Could not resolve 'security.ubuntu.com'  
Err http://us.archive.ubuntu.com trusty InRelease  
Err http://us.archive.ubuntu.com trusty-updates InRelease  
Err http://us.archive.ubuntu.com trusty-backports InRelease  
Err http://us.archive.ubuntu.com trusty Release.gpg
```

Figure 9: Final escalation validation sequence displaying administrative capture (8.jpeg).

11. GitHub Repository Layout

The tracking files and solution configurations are submitted under the following structural framework layout [cite: 5]:

```
ashwini-k-ctf-appliance/  
├─ README.md # Operational configuration walkthrough guidelines  
[cite: 5]  
├─ VM_Appliance/  
│ └─ Ashwini_K_CTF_Machine.zip # Comprised .ova virtual distribution target  
[cite: 5]  
├─ Web_Source/  
│ └─ index.html # Core portfolio site layout [cite: 5]  
│ └─ assets/css/main.css # Styling configuration including stego passkey  
hints [cite: 5]  
└─ Documentation/  
└─ CTF_Development_Report.pdf # Engineering document publication target [cite:  
5]
```

12. Conclusion

The custom Linux CTF appliance running under the hostname **Ashwini K** has been successfully built, verified, and exported [cite: 5]. It fulfills all technical guidelines by testing multi-stage evaluation processes, web security parameters, hidden directory enumeration, client data extraction, and administrative escalation [cite: 5].